



PORTFOLIO

Ahilan Kalirajah

A piece of my design work





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ABOUT ME

Mechanical engineering graduate with strong research, continuous improvement strategies and hand-on-skills. Has been involved in academic and professional projects which prioritize management, R&D, teamwork, reporting and technical communication.



EXPERIENCE

01

November 2022 - Present
Methods Engineering Specialist
Safran

02

July 2021 - Present
Junior Plant Engineer
North American Steel

03

June 2020 – July 2021
Mechanical Technologist and CNC programmer
Misani Custom Design

04

July 2018 – August 2018
Press Setup Technician
Peel Plastic Products Ltd.



EXPERIENCE

04

September 2016 – April 2017
Capstone Project
Custommachinery INC.

June 2016 – August 2016
Robotic Welding Machine Operator
Matcor Metal Fabrication

05



EDUCATION

January 2015 – April 2017

Advanced Diploma in Mechanical Engineering Technology – Design
George Brown College

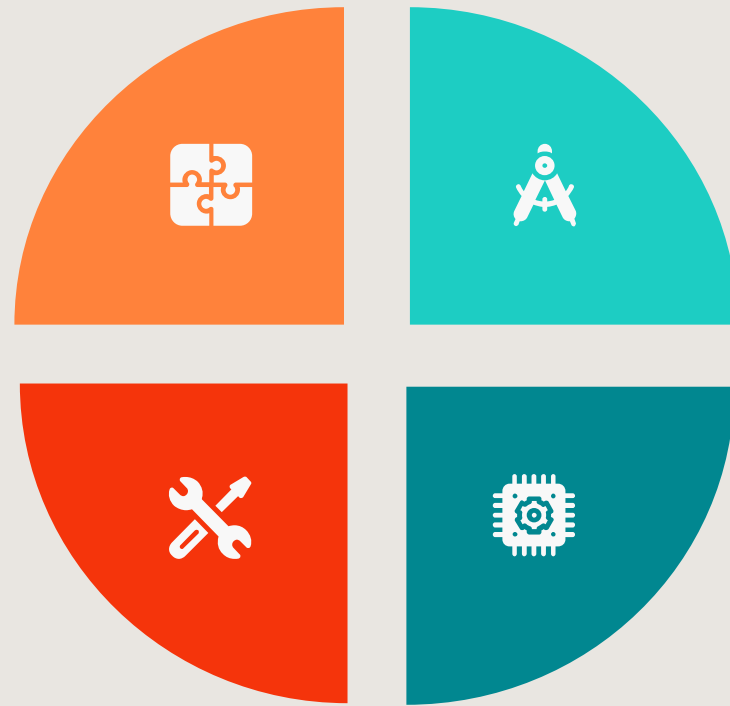
01

02

September 2017 – April 2020

Bachelors Degree of Engineering in Mechanical Engineering
Lakehead University





Work Projects



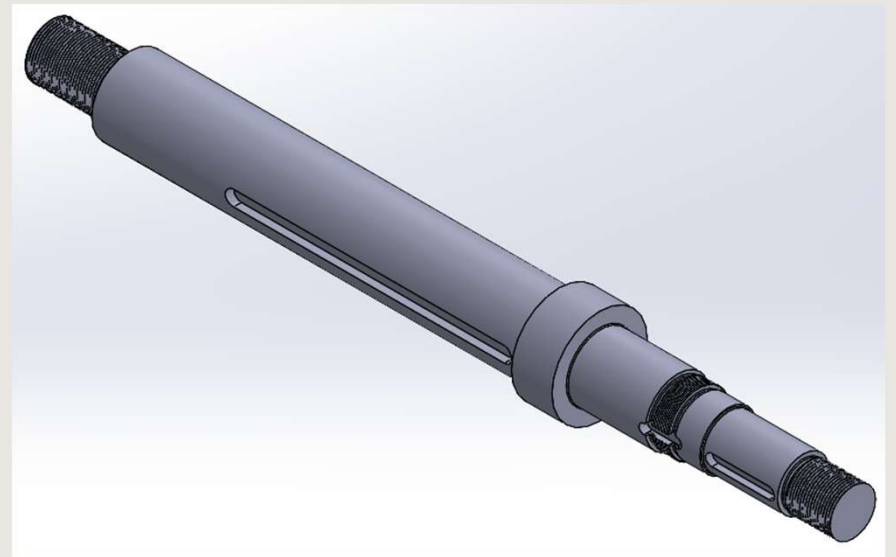
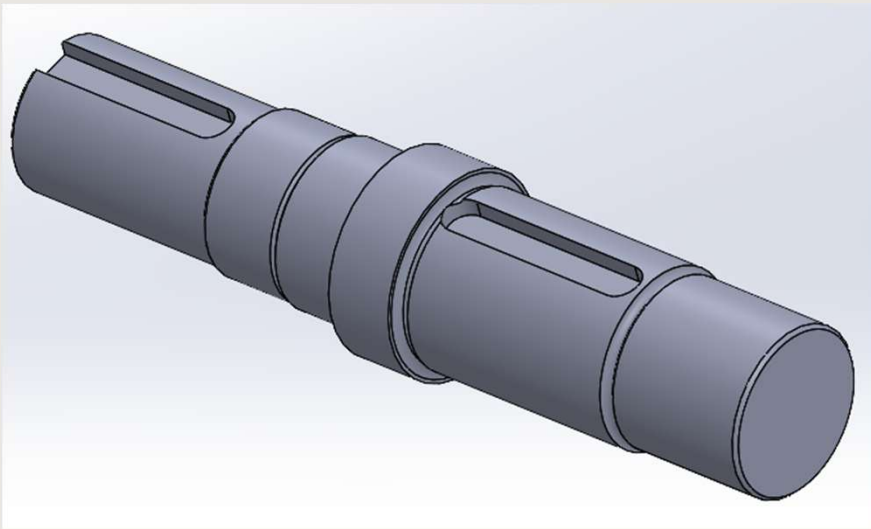
Solidworks

North American Steel 

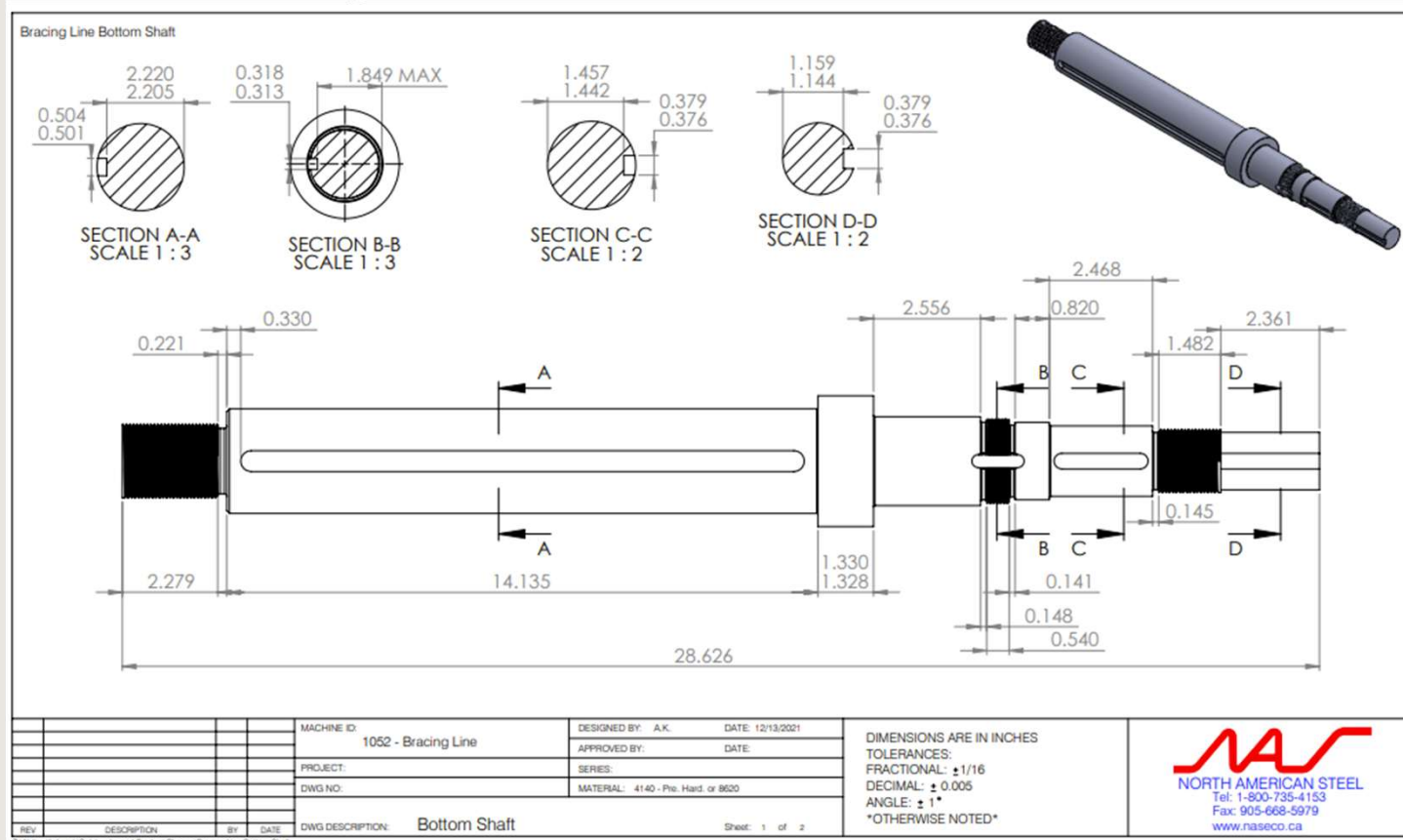
North American Steel is pallet racking and industrial storage manufacturing company. Working here one of my main tasks was to create Solidworks models of the parts we were manufacturing. The following slides show some of the parts and assemblies which I was tasked with converting to Solidworks



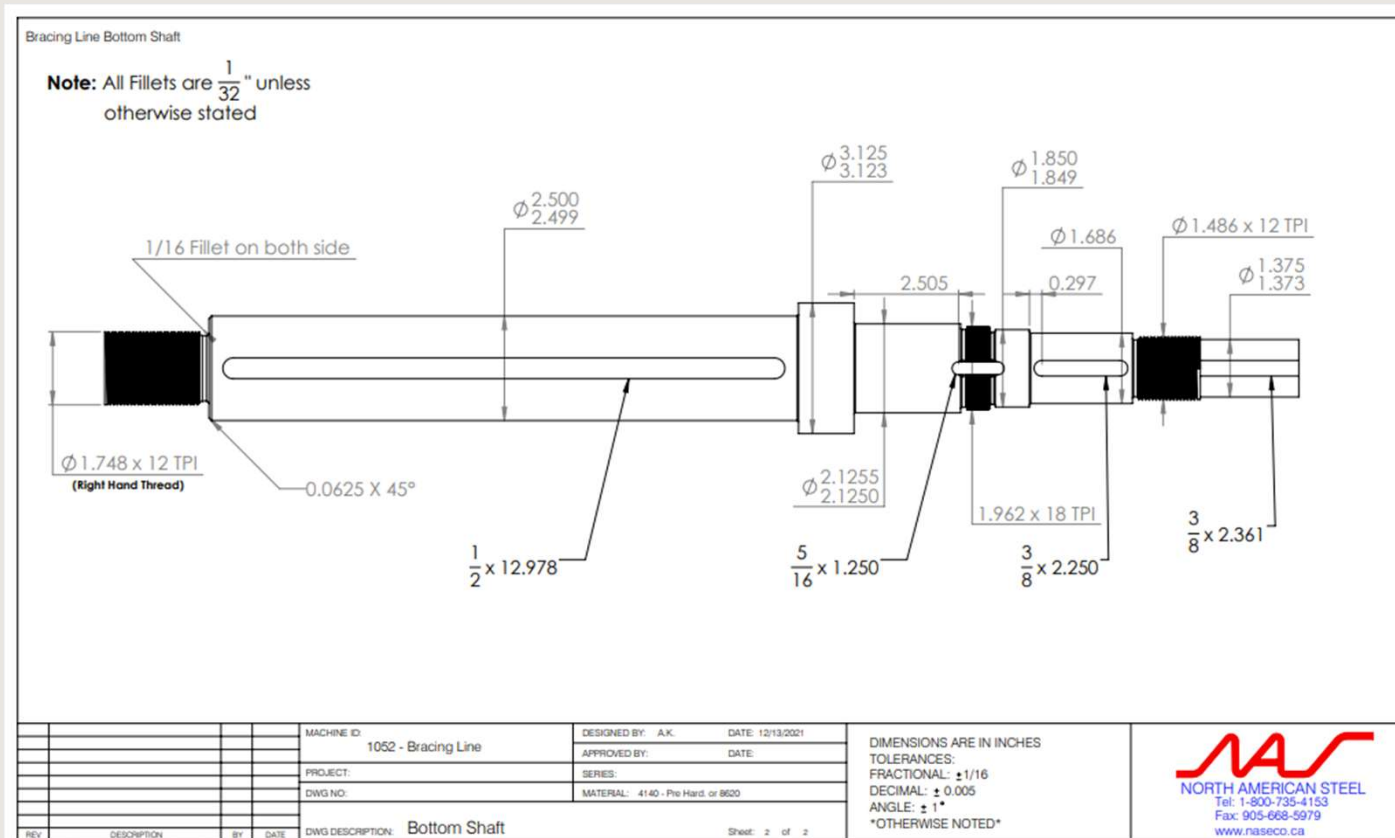
Shafts



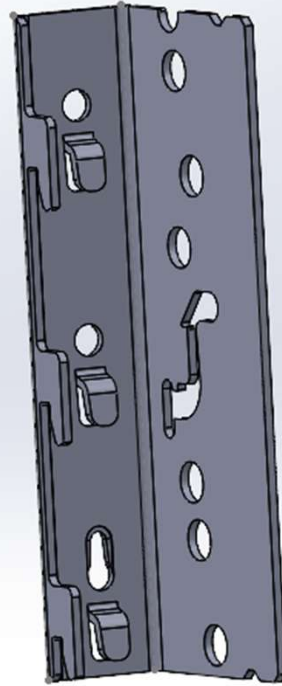
Shafts - Drawing



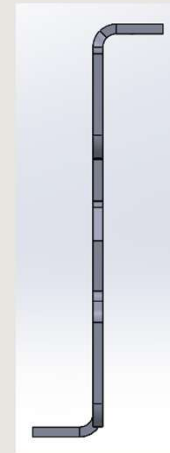
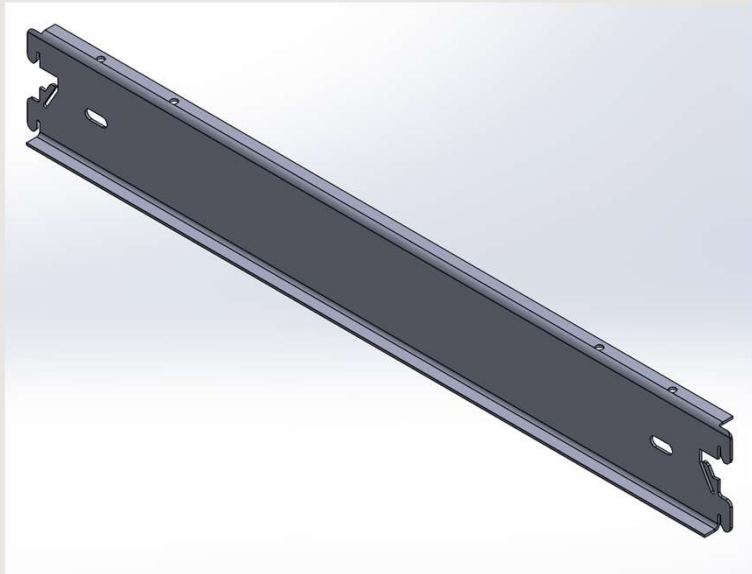
Shafts - Drawing



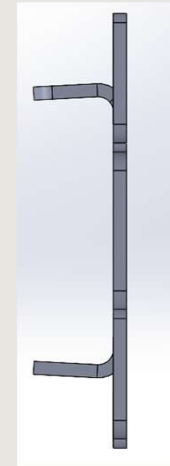
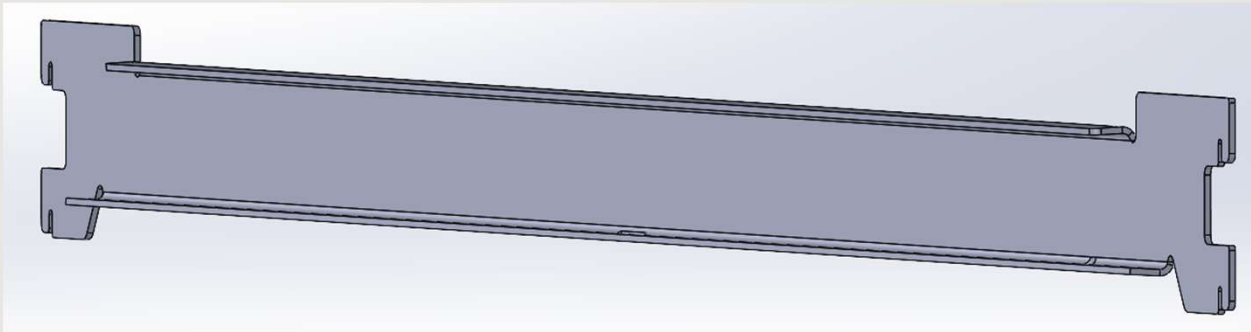
Step-Up Beam Plate



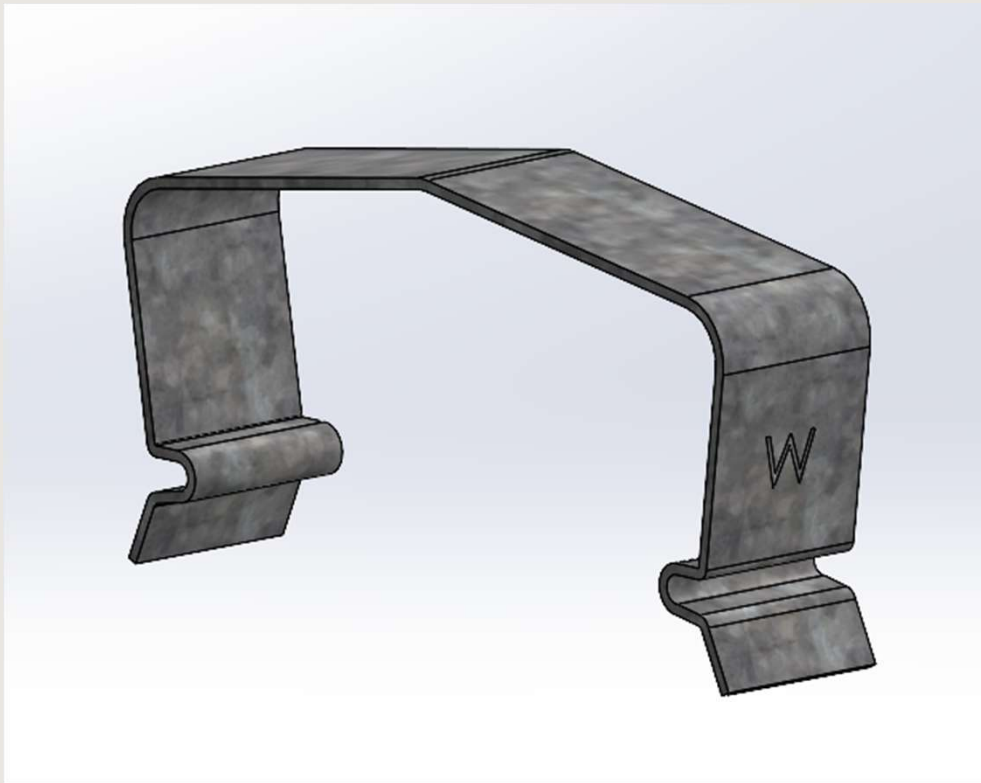
Cross-Member Part



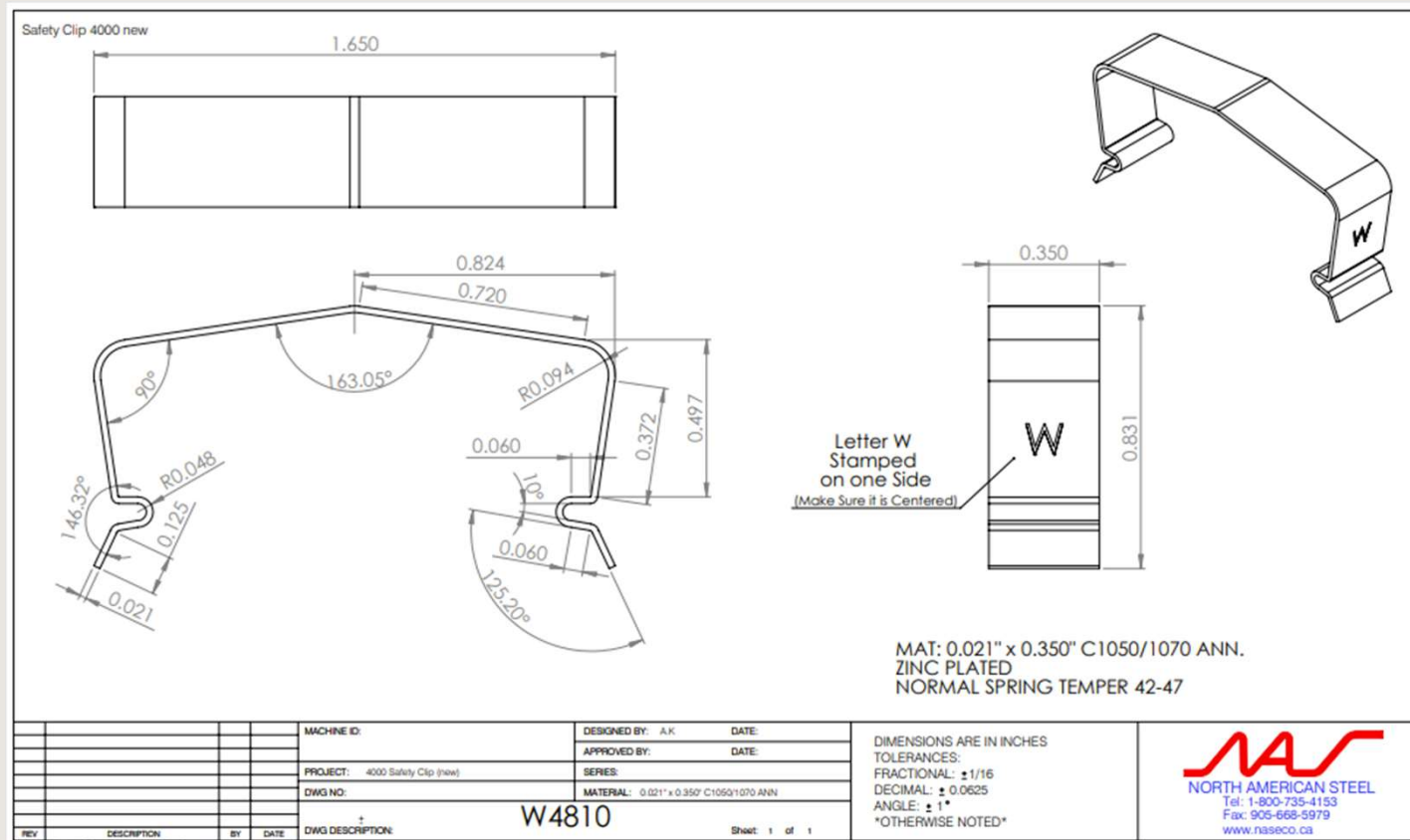
Medium-Crossmember Part



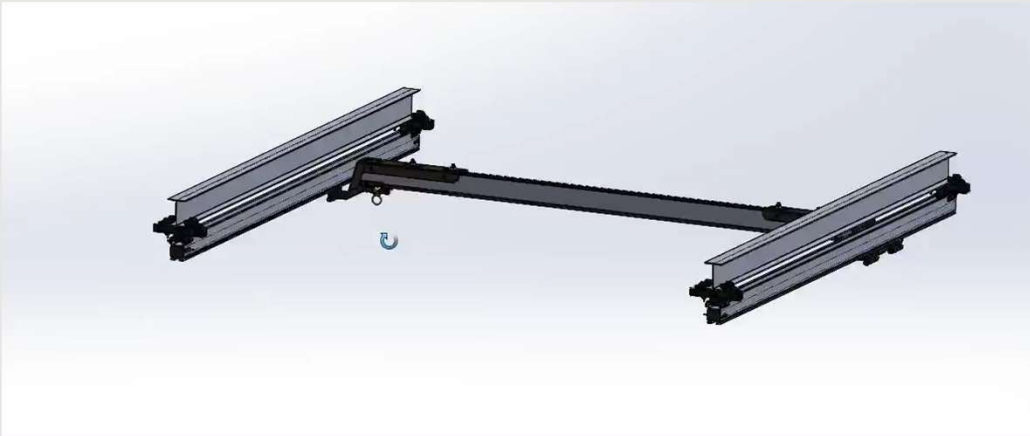
Safety Clip



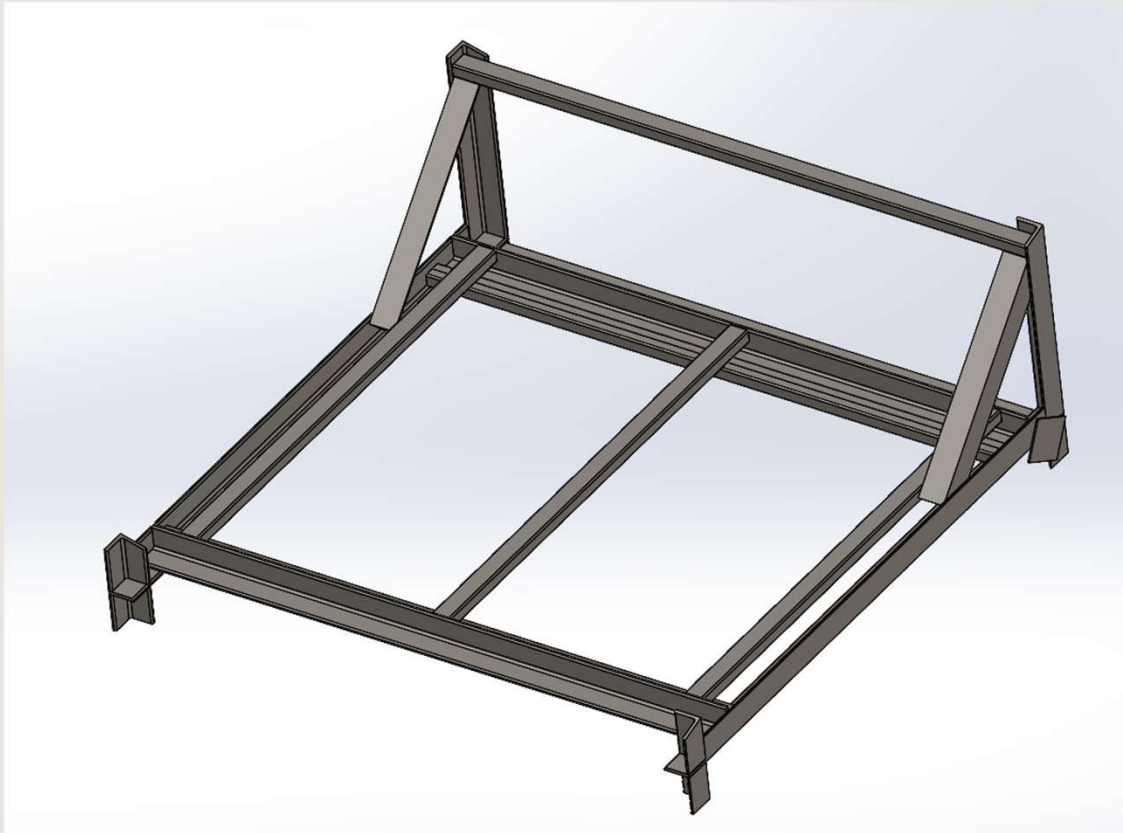
Safety Clip - Drawing



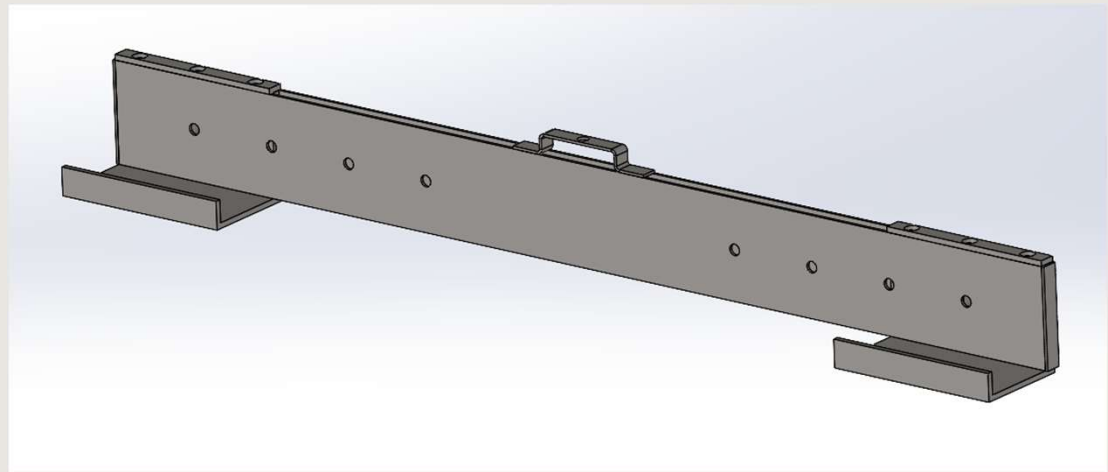
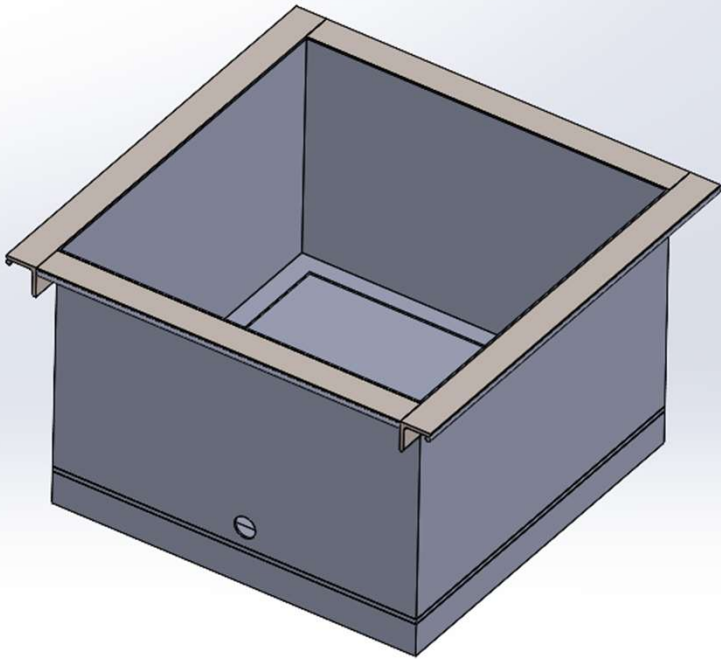
Paint Line Lift



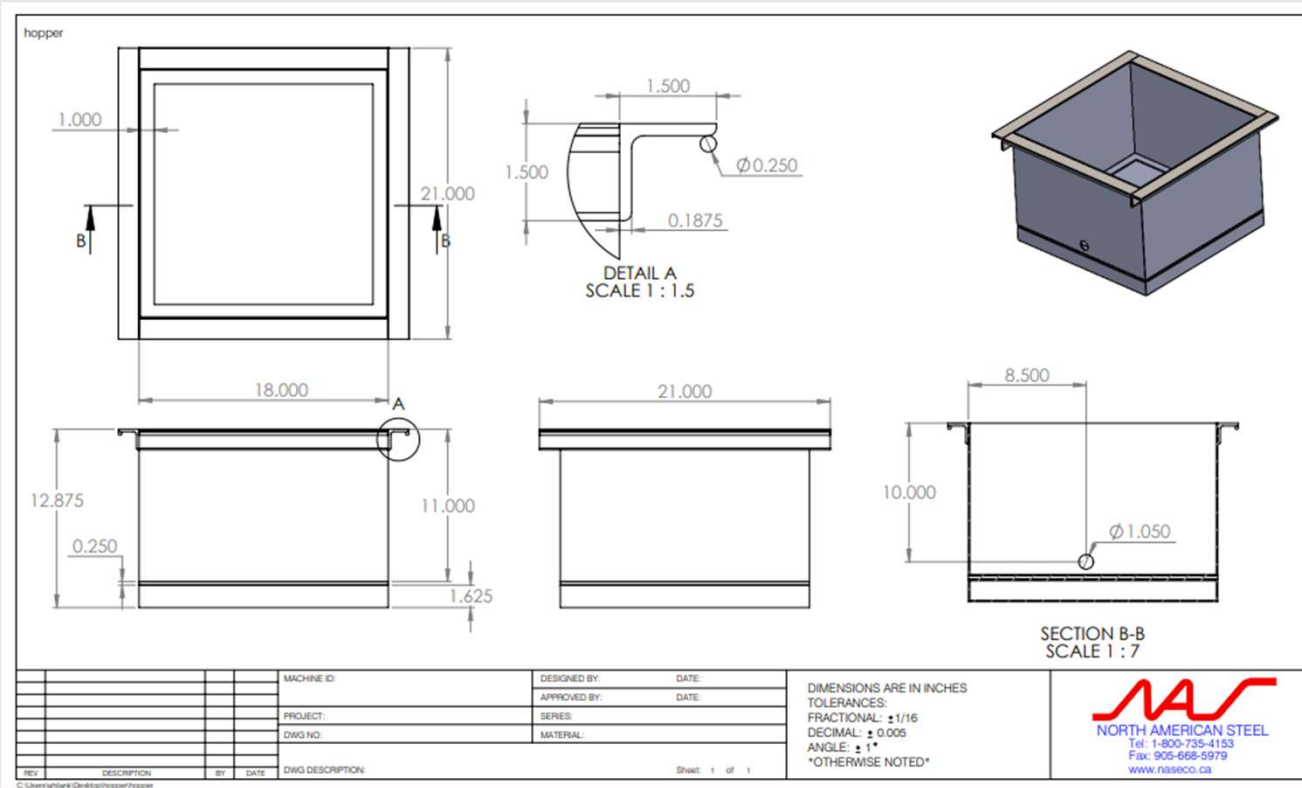
Basket Holder Assembly



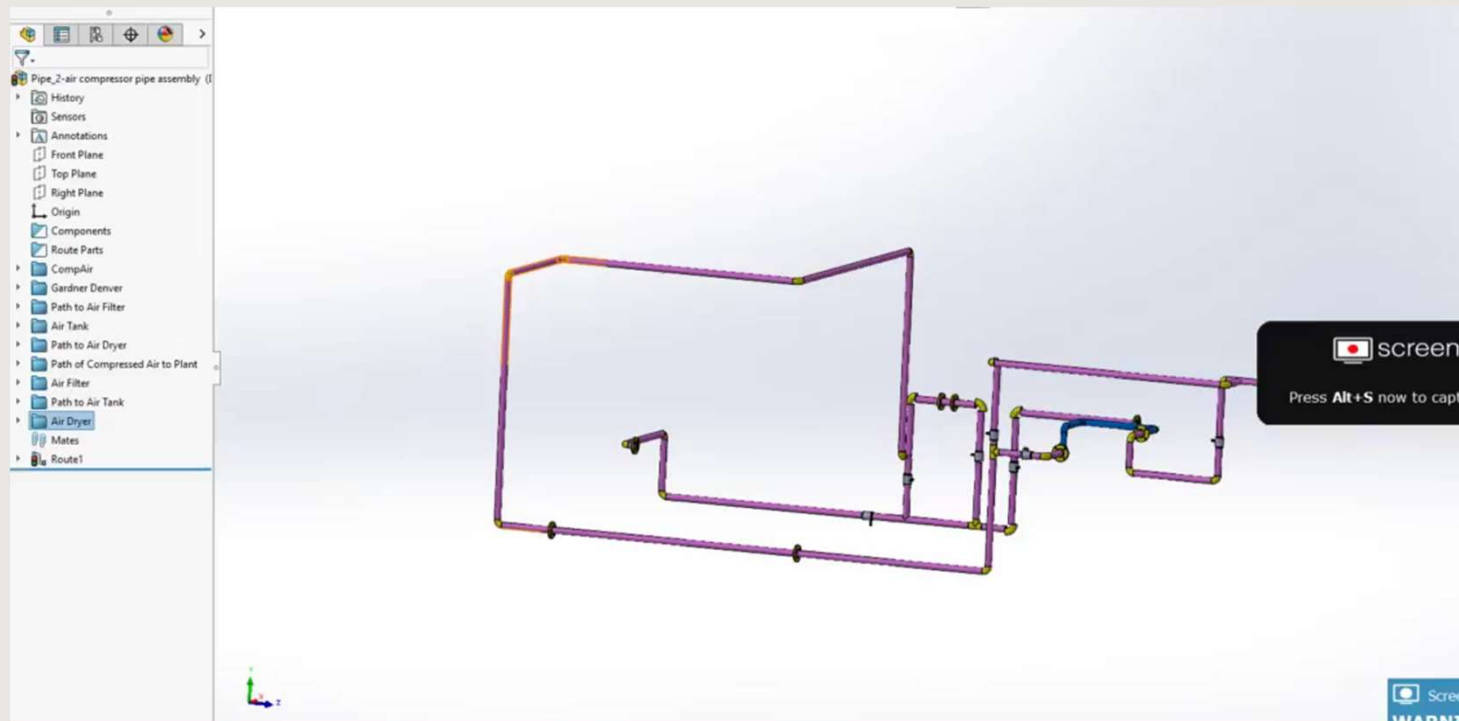
Paint Line Hopper and Paint Line end-effector Assembly



Paint Line Hopper Drawing



Compressor Room Project Piping Layout



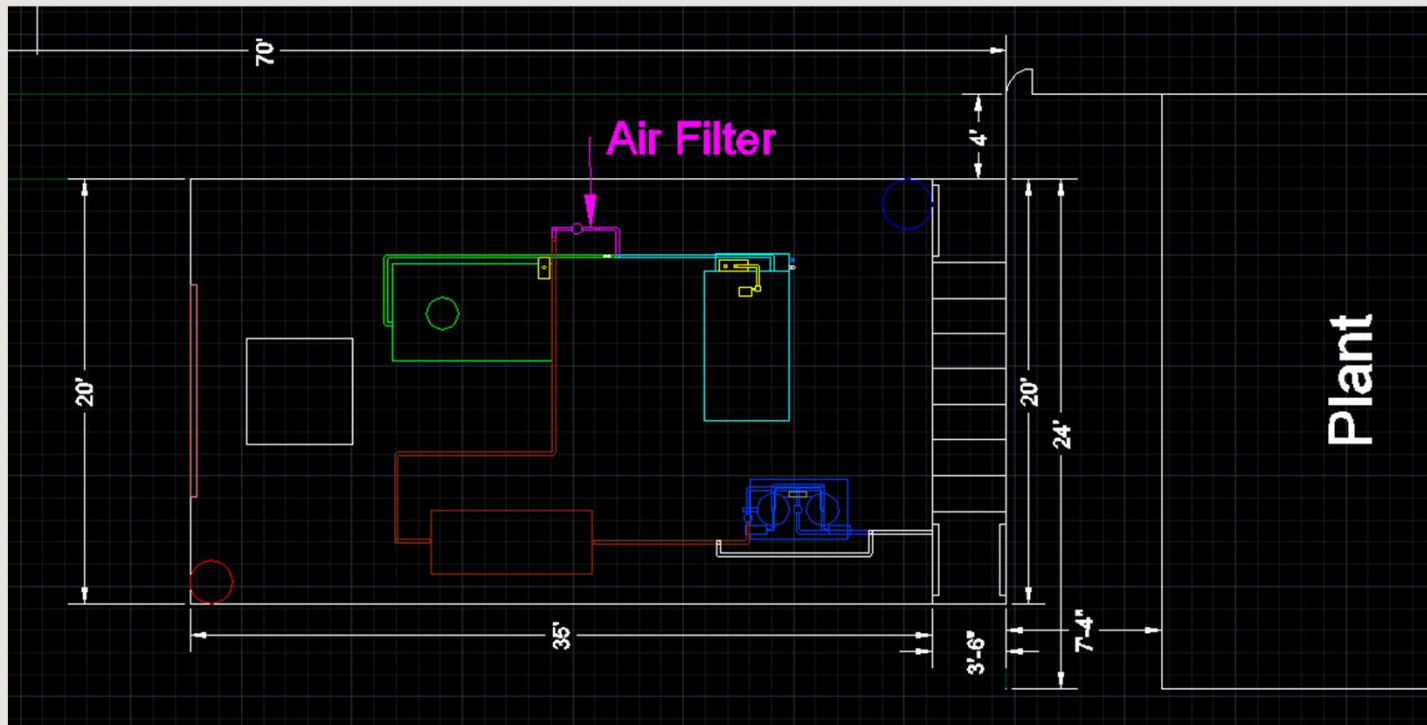
AutoCAD – Compressor Room Project

North American Steel 

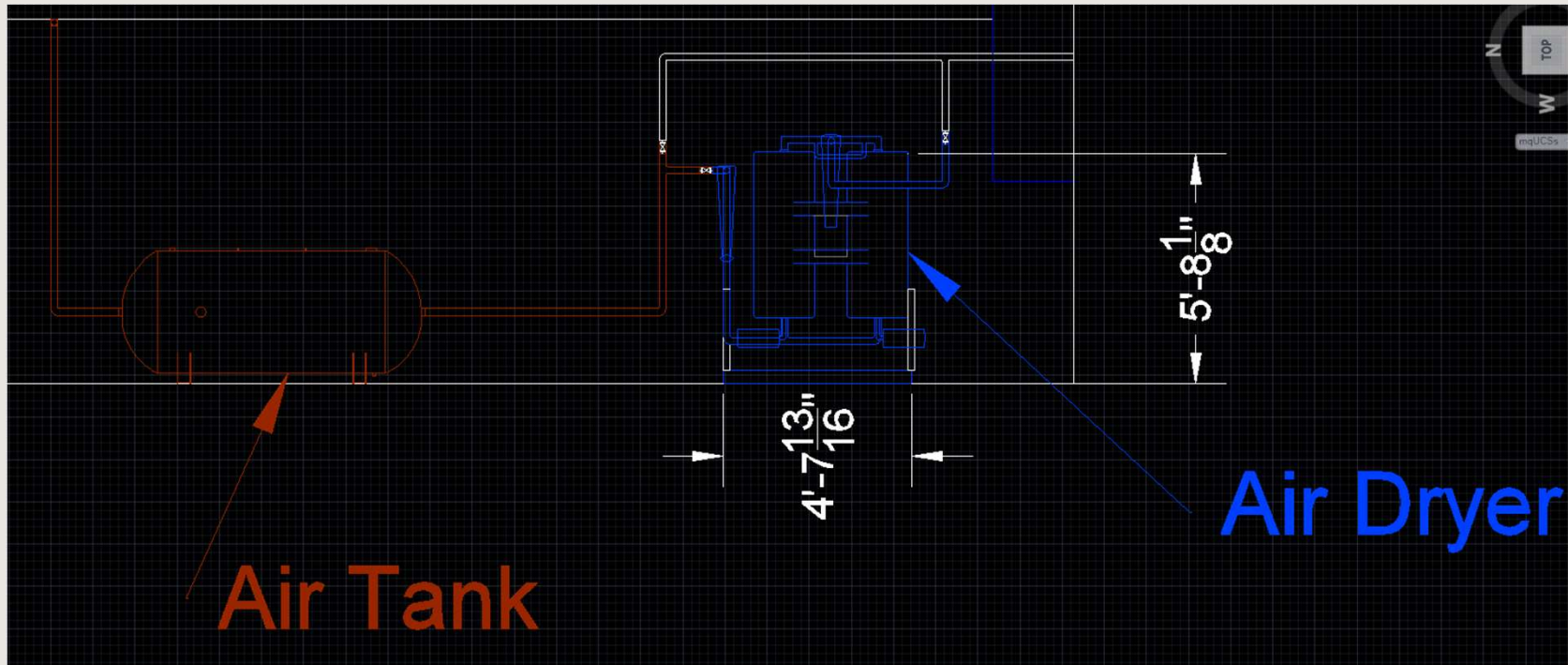
The purpose of this project was to create a compressor room located right outside the plant since they are currently located in a location that is not allowing it to run efficiently. I was tasked with modelling our two compressors, air dryer and tank out on AutoCAD and then placing them in a room and creating a sample layout. This layout was used to give an idea of what we were looking for when prospecting potential contractors.



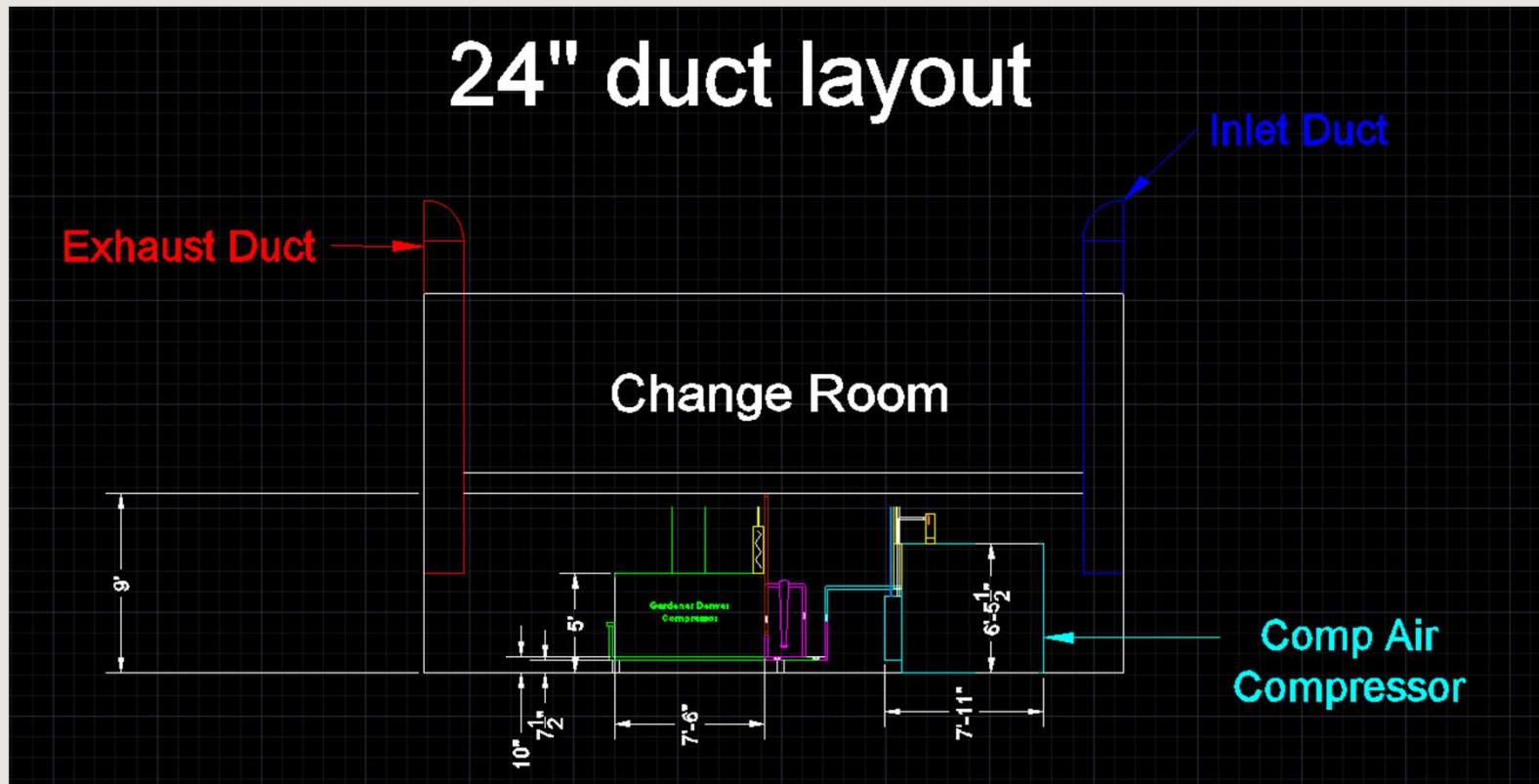
AutoCAD Compressor Room Layout (Top View)



AutoCAD Compressor Room Layout (Front View)



AutoCAD Compressor Room Layout (Front View)



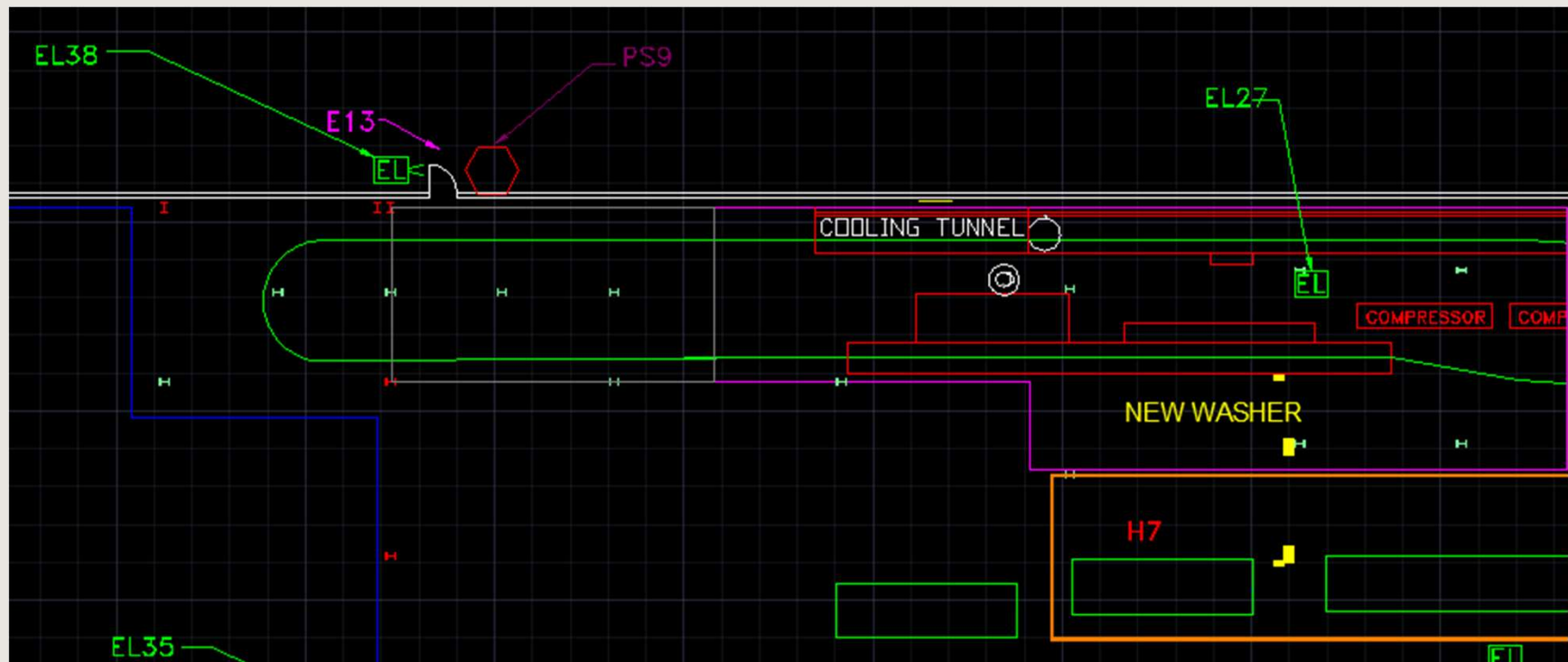
AutoCAD – Paint Line Lift Project

North American Steel 

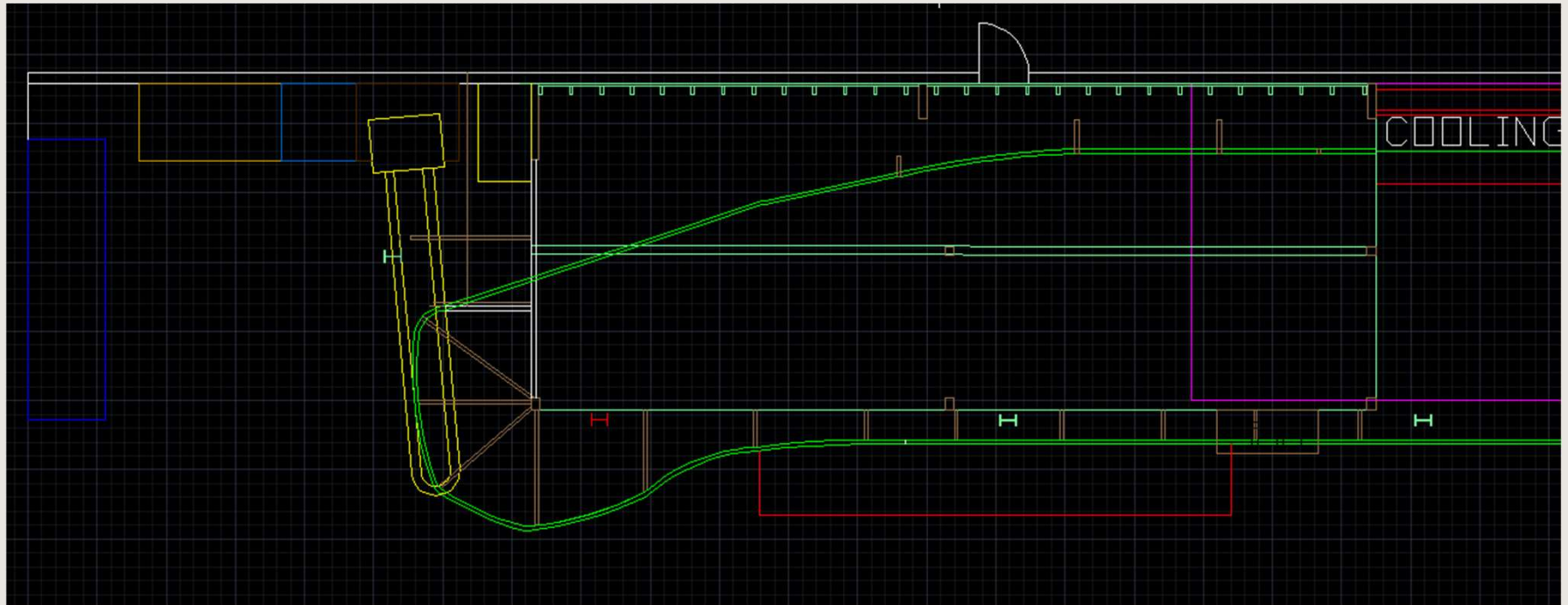
The purpose of this project was to create a lifting device which would assist the workers to unload the finished frames on the unloading side of the paint line. Since the unloading side was located under a mezzanine the workers would be stuck taking these frames off by hand which was dangerous since these frames ranged from 200 - 600 lbs and ranged from being 8' - 31' long. Before I could start coming up with a solution to the problem I needed to have an accurate drawing of what the paint line looked like from the top, side and front views. The only drawing we had was from our current plant layout which was an only a top view drawing and it looked nothing like what the current paint line looked like. So for the first 1-2 months of the project I drafted out the paint line on AutoCAD.



AutoCAD Paint Line Layout (Top View Old Layout)

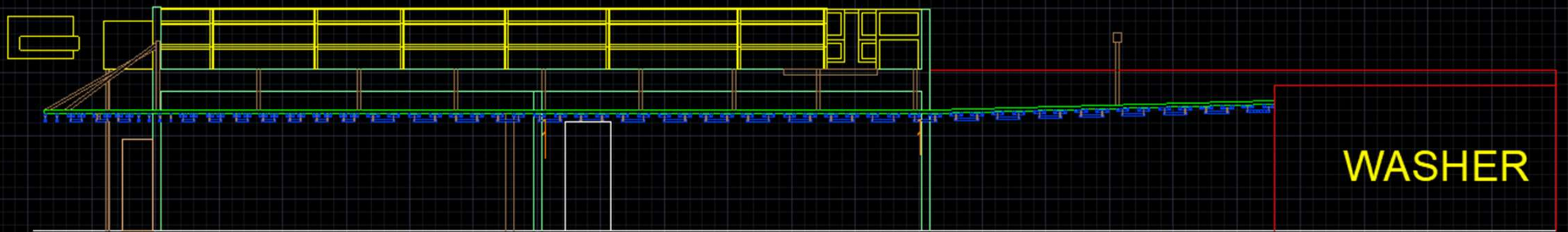


AutoCAD Paint Line Layout (Top View Updated Layout)



AutoCAD Paint Line Layout (Front View)

Front View

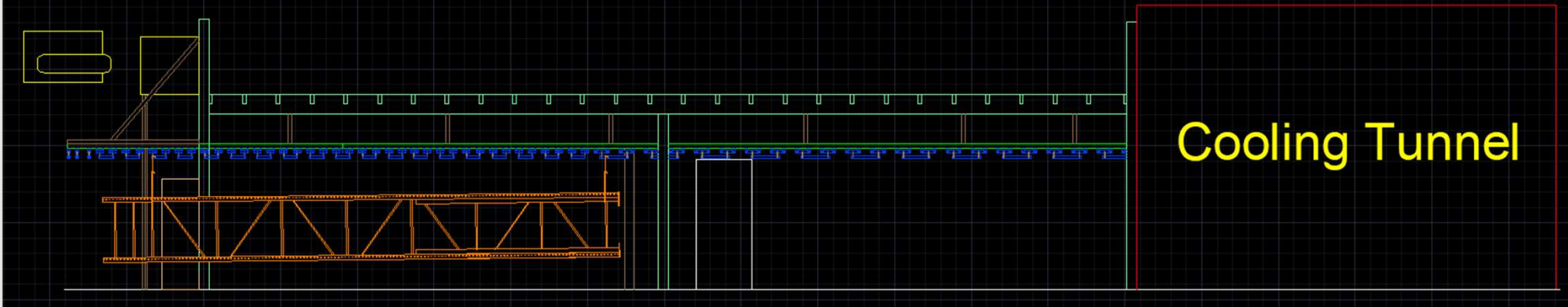


AutoCAD Paint Line Layout (Cross - Sectional View)

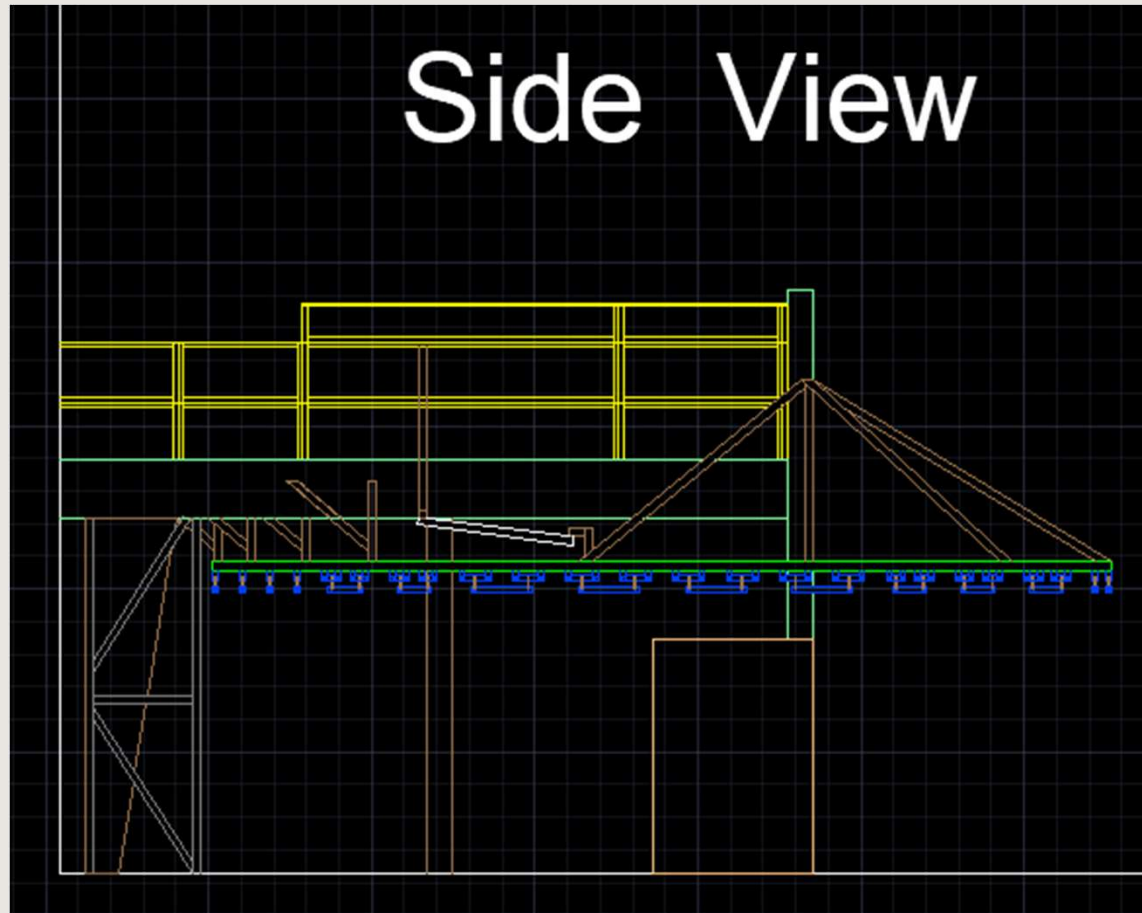


AutoCAD Paint Line Layout (Front View With Beam)

Cross-Sectional View With Beam



AutoCAD Paint Line Layout (Side View)



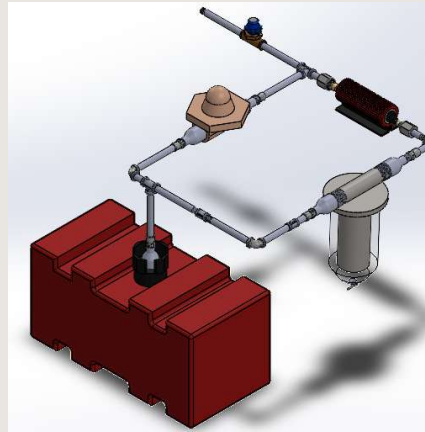
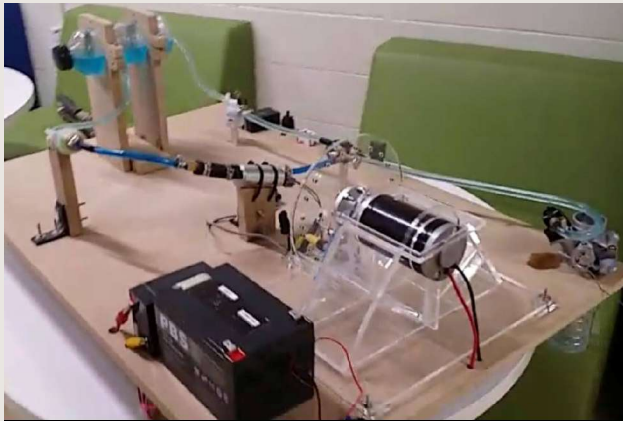


Design Projects

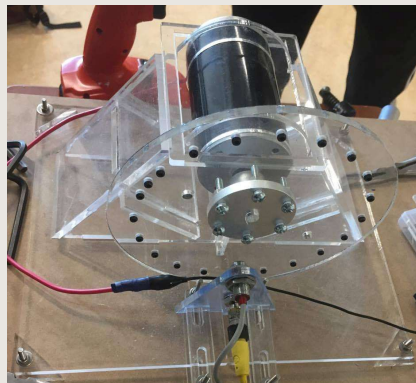
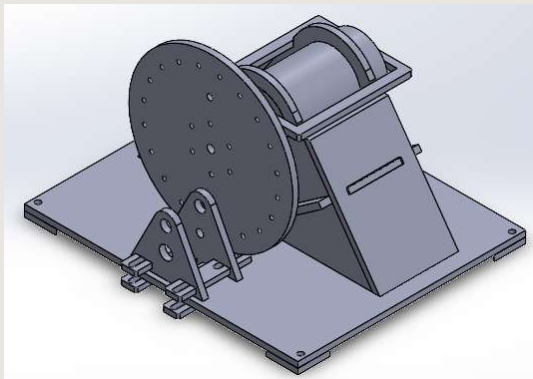


Custommachinery INC. – Fuel Injection System

Capstone Project - George Brown College



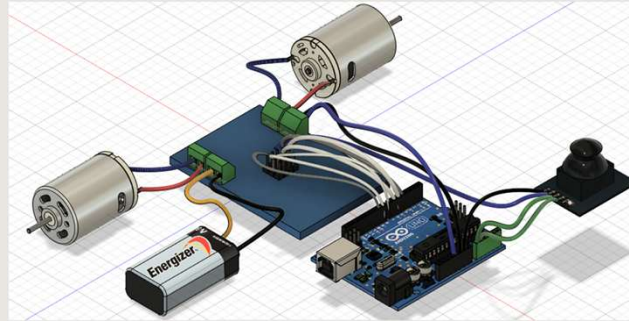
For my capstone project the client that I chose to work for was working on creating a new type of rotary engine and needed a working fuel injection system which he could use to test different types of fuel mixtures. We were successful in being able to reduce the cost of his system by 60%. My role was to create all of the Solidworks models for both our prototype and final models. I was also in charge of using Mastercam X9 and Solidworks to laser cut and CNC some of the components which we used in our final model.



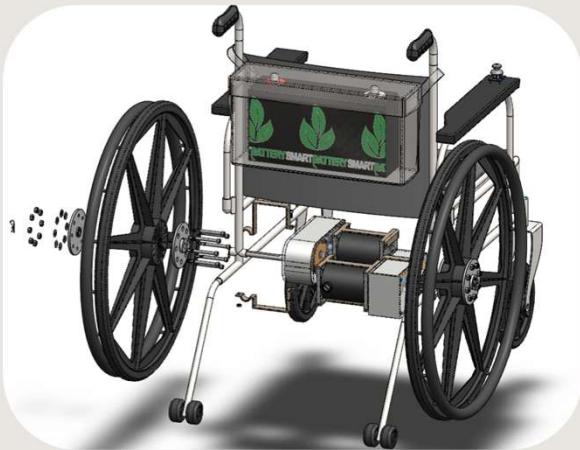
Manual to Electric Wheelchair Attachment

Capstone Project – Lakehead University

[Electric System Proof of Concept Link](#)



My capstone project was to try to convert a manual wheelchair into a electric one. We were successful in being able to create a prototype model which was 75% cheaper compared to the average electric wheelchair and achieved a easy to install system. My main role in this project was to come up with a electric system which would move the wheelchair front, back, left and right. I used Fusion 360 to create the CAD model of the electric system. I also assisted with creating the Solidworks components for the prototype wheelchair attachment.

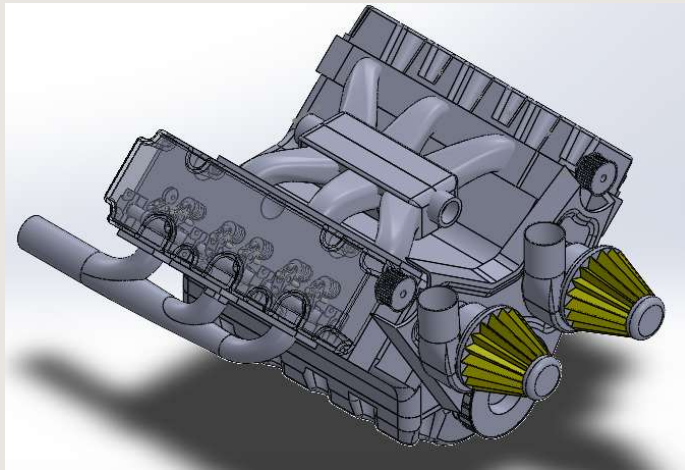


[Solidworks Animation link](#)

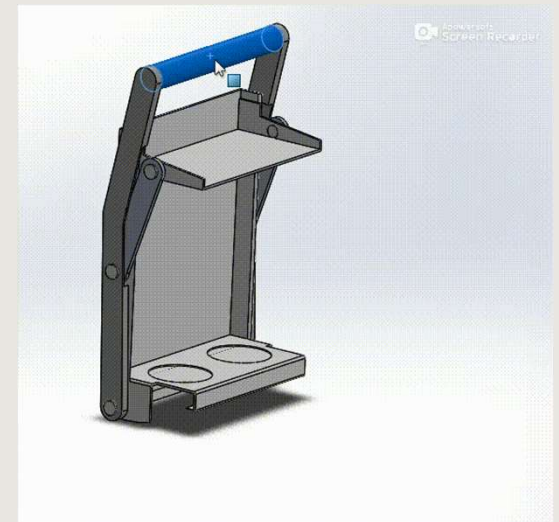


Hobby Projects

Gasoline Engine Concept



Can Crusher Concept

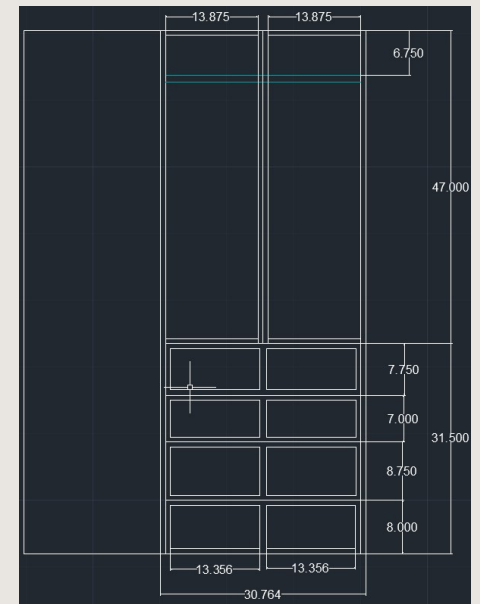
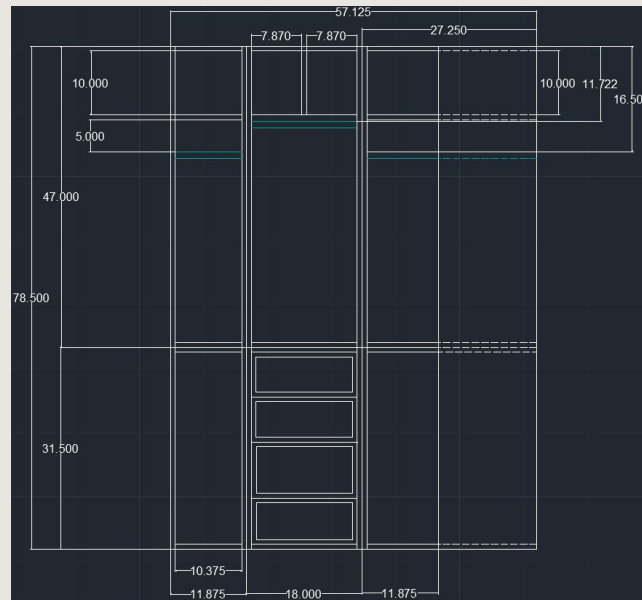
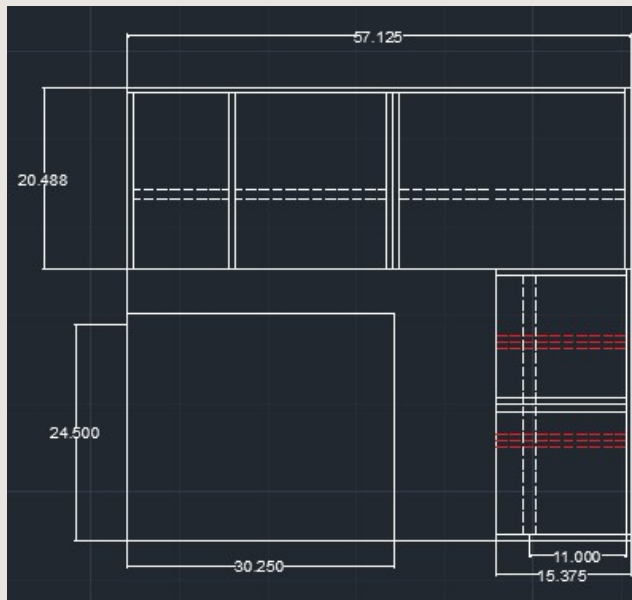


- For both these assemblies I created individual parts for each of the components used in the assembly
- I used different types of mates ranging from standard mates to mechanical mates



Hobby Projects

Misani Custom Design - Walk-In Closet Design



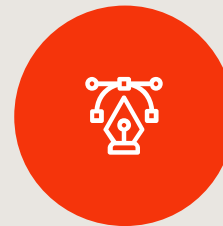
- This is a drawing which I made using AutoCAD which represents the design for a walk-in closet which I am planning to make for my house.
- This is the finalized design which I have recently finished which shows the top view and front views of the cabinets



COMPUTER SKILLS



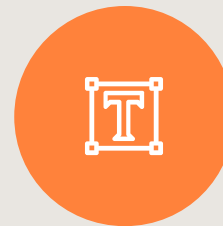
AutoCAD LT



Solidworks



Mastercam



Microsoft Office



COMMUNICATION SKILLS



COMMUNICATION SKILLS

Mother tongue(s)

- English
- French

Profficiency

English ● ● ● ● ● ●

French ● ● ● ● ● ○

